UI/UX DESIGN SPECIFICATION • DESIGN SYSTEM V1.0

# MemoryVault AI: UI/UX Specification

Complete Design Blueprint: User Journey, Conversational Triage, Route Visualization & Design System Tokens

| DESIGN FOCUS | PRIMARY PATTERN | ACCESSIBILITY | FORM FACTOR |
|---|---|---|---|
| Cognitive De-escalation | Hybrid Chat + Live Spatial Card | WCAG 2.1 AA Compliant | Responsive Web (Mobile-First) |

## 1. UX Philosophy & Core Design Principles

When people misplace everyday essentials (keys, wallets, earphones), they are usually experiencing acute cognitive anxiety, rushed schedules, or distraction. The interface must act as a calming cognitive anchor rather than an intimidating form or cluttered dashboard.

**1. Anxiety Reduction & Immediate Clarity**
Prioritize serene typography, non-alarming cool colors, and instant feedback. Eliminate dead ends and complex configuration menus.

**2. Conversational Efficiency (Anti-Fatigue)**
Users in a hurry cannot fill 10 form fields. The chatbot limits clarification to 1–2 rapid interactive chip taps before generating a search plan.

**3. Actionable Spatial Scaffolding**
Raw probabilities (e.g. "Kitchen: 62%") induce hesitation. The UI translates statistical predictions into a physical, step-by-step walking checklist.

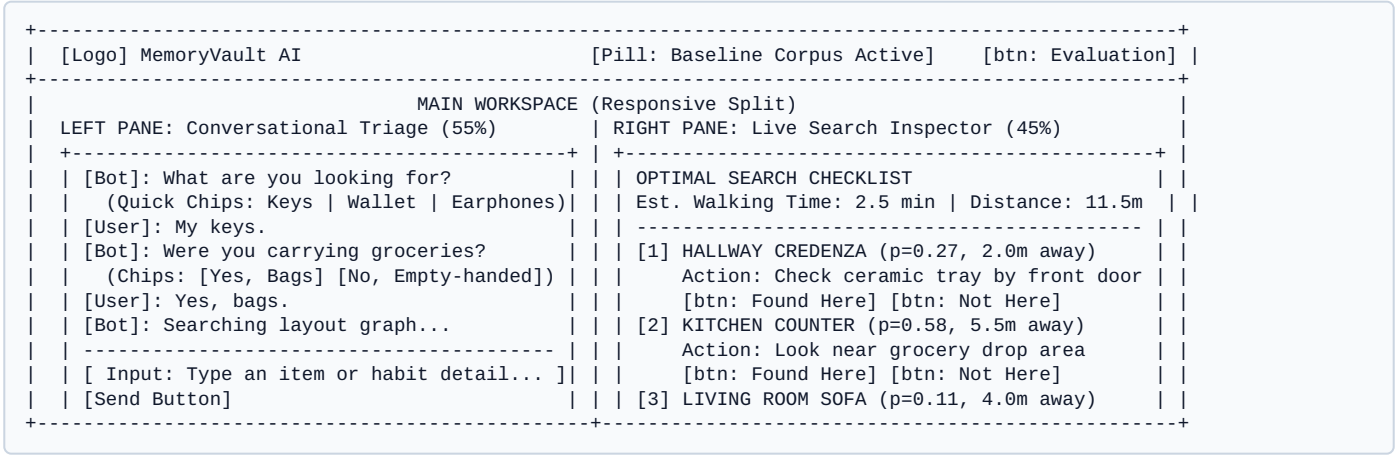
**4. One-Tap Feedback & Micro-Closure**
Finding the item brings instant relief; ticking "Found Here!" provides emotional closure while transparently reinforcing the underlying model.

## 2. End-to-End User Journey & Mental Model

| Stage | User Mindset & Goal | System Touchpoint & Interaction | UX Success State |
|---|---|---|---|
| **1. Trigger & Panic** | "I can't leave the house without my keys. Where did I leave them?!" | User opens MemoryVault web app. Instant focus lands on a clean input bar with preset item pills. | Frictionless entry in < 3 seconds without sign-in barriers. |
| **2. Clarification** | "I don't remember details, but I recall it was raining heavily." | Chatbot asks: *"When did you arrive, and did you have bags?"* Dynamic multi-choice chips render for 1-tap answering. | Context vector established with zero manual typing required. |
| **3. Spatial Triage** | "Show me exactly where to walk first so I don't run around." | Search Route Card slides into view: Step 1 (Hallway Bowl, 2m) → Step 2 (Kitchen Island, 5m). Steps show micro-actions. | User immediately begins physical search along the optimal trajectory. |
| **4. Resolution** | "Found them in the grocery sack on the counter!" | User taps ✓ `Found Here!` on Step 2. Affirmative confetti/ripple effect displays; feedback logs to SQLite. | Closure achieved; user proceeds to leave on time with gratitude. |

## 3. Information Architecture & Navigation Structure

MemoryVault AI operates as a responsive Single-Page Application (SPA) structured into two unified panes:




## 4. Detailed Screen Specifications & States

### Screen 1: Primary Conversational Workspace

- **Top Utility Bar:** Minimalist brand identity, system readiness status pill (*"Model Ready • 5,000 priors"*), and an Evaluator Analytics toggle for academic demonstration.

- **Chat Stream:** Clean vertical bubble thread. System messages appear with soft grey/blue cards with high contrast; user messages are right-aligned in brand primary blue.
- **Adaptive Context Quick-Chips:** Horizontal scrolling pill buttons dynamically populated below bot queries so users can answer with a single thumb tap (e.g., `[Heavy Rain]`, `[Carrying Groceries]`, `[Past Midnight]`).
- **Omni-Input Bar:** Auto-expanding text field with placeholder *"Ask where you left keys, wallet, etc..."*, voice-disabled indicator, and primary send icon.

### Screen 2: Live Search Plan & Route Inspector

- **Route Summary Header:** Displays total computed physical transit distance (meters), estimated inspection time (minutes), and dominant habit cluster tag (e.g., *"Cluster #1: Rushed Grocery Return"*).
- **Step Cards:** Ordered sequential checklist generated via A* pathing. Each card features:
  - **Step Number Badge:** Highlighted primary circle (1, 2, 3).
  - **Candidate Zone Title:** Clear room & furniture designation (e.g., *Kitchen Counter Island*).
  - **Confidence Bar & Probability Tag:** Micro progress meter showing individual zone posterior likelihood.
  - **Specific Inspection Action:** Contextual prompt (e.g., *"Inspect surfaces next to grocery bags"*).
  - **Inline Outcome Buttons:** Prominent `Found Here!` (Emerald Green) and `Not Here` (Ghost Slate).

### Screen 3: Evaluator Analytics Drawer (Modal / Slide-Over)

Targeted at examiners, evaluators, and system developers. Slides from the right when clicking `[Evaluation Metrics]`:

- **Confusion Matrix Heatmap:** Tabular representation of true vs. predicted household drop locations.
- **ROC-AUC Metric Card:** Visual display of One-vs-Rest ROC curve and macro-average score (≥ 0.85).
- **PCA 2D Cluster Projection:** Interactive scatter visualization of the ~5,000 synthetic records grouped by K-Means into behavioral patterns.

## 5. Interaction States, Loading, Empty & Edge Handling

| State Type | UI / Sensory Manifestation | UX Objective & Recovery Path |
|---|---|---|
| **Empty State** | Hero greeting card: *"Misplaced something? Let's retrace your steps."* with 3 starter chips (`Keys`, `Wallet`, `Earphones`). | Eliminates the "blank canvas" freeze. Prompts immediate action with single-tap starters. |
| **Loading / Inferring** | Three bouncing dots in chat + subtle skeleton loader in the Route pane with caption: *"Calculating optimal A* walking path..."* | Assures user computation is active. Maximum 1.5s duration prevents frustration. |
| **High Entropy Warning** (Low Confidence) | Soft amber banner: *"Unusual routine detected. Top locations are scattered. Here is our best conservative check order."* | Manages expectations honestly; avoids false precision while still providing a structured walking order. |
| **Unrecognized Item** | Assistant response: *"I can't reliably track that item yet. Supported items: Keys, Wallet, Earphones, Glasses, Bag."* | Immediate clarity without cryptic error codes; surfaces chips of supported items. |
| **Success State** | Celebratory green highlight on the card, subtle sound/haptic pop, and confirmation message: *"Great! Habit logged for next time."* | Delivers immediate relief and closure, rewarding the user for completing the cycle. |

## 6. Design System Tokens: Colors, Typography & Spacing

### 6.1 Color Palette

| Token Name | Hex Value | Preview | Functional Application |
|---|---|---|---|
| `--color-brand-primary` | `#0284c7` | Sky 600 | Primary buttons, active step badges, user chat bubbles. |
| `--color-brand-dark` | `#0f172a` | Slate 900 | Header background, primary headings, terminal code blocks. |
| `--color-bg-base` | `#f8fafc` | Slate 50 | Main app canvas and card inset backgrounds. |
| `--color-surface-card` | `#ffffff` | Pure White | Route cards, bot chat bubbles, evaluation containers. |
| `--color-success` | `#059669` | Emerald 600 | "Found Here" confirmation buttons, high confidence metrics. |
| `--color-warning` | `#d97706` | Amber 600 | Low-certainty indicators, unusual routine notifications. |
| `--color-border` | `#e2e8f0` | Slate 200 | Subtle card borders, divider lines, disabled input rings. |

### 6.2 Typography Hierarchy

- **Font Family:** `-apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, "Helvetica Neue", sans-serif` (Native system font stack for zero download latency).
- **Page Title:** 18pt / Line-height 1.2 / Bold 800 (Clean, commanding brand banner).
- **Section Headers (H2):** 11pt / Line-height 1.25 / Bold 700 with primary accent bar.
- **Component Titles (H3):** 9.2pt / Semi-Bold 600 / Slate 900.
- **Body Text:** 8.5pt / Line-height 1.45 / Regular 400 / Slate 700.
- **Micro-Labels & Metadata:** 7.0pt / Uppercase / Letter-spacing 0.5px / Bold 700.

### 6.3 Spacing & Layout Grid

Strict 8pt Spatial Grid: `4px (micro)`, `8px (compact)`, `16px (standard)`, `24px (loose)`, `32px (layout gutters)`. Touch targets on interactive chips and buttons are guaranteed to have a minimum dimension of **44 × 44px** for mobile accessibility.

## 7. Responsive Behavior & Breakpoints

**Desktop (> 1024px)**
Dual-column side-by-side layout. Left pane (55%) dedicated to conversation; right pane (45%) hosts the live-updating search route checklist and interactive mini-map.

**Mobile (< 768px)**
Single-column view. Focus stays on the chat stream; upon route generation, a bottom sheet card slides up with a swipe indicator, allowing full route exploration without losing chat history.

## 8. Accessibility (a11y) & Usability Standards

- **Color Contrast Ratios:** All body text achieves a minimum contrast ratio of **6.5:1** against light backgrounds, far exceeding the WCAG 2.1 AA requirement (4.5:1).
- **Keyboard Navigability:** Full keyboard support (`Tab`, `Shift+Tab`, `Enter`, `Space`). Quick-chips and action buttons feature prominent `:focus-visible` rings.
- **Screen Reader (ARIA) Annotations:** The chat log is declared as an `aria-live="polite"` region so new clarification questions and route cards are announced automatically without disrupting user focus.
- **Cognitive Guardrails:** Jargon like "Multiclass Random Forest" and "A* Search Heuristic" is hidden from the primary consumer view and isolated inside the Evaluator Drawer.